

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Allow 25.4/ 25.6 / 26.0/ 26.3/ 26.25µm for 2 marks Allow 1 mark for (104/20)x 5 or (104/20.5) x 5 or (105/20) x 5 or (105/20.5) x 5 104000/ 4000 or 105000/ 4000 or 104000/ 4100 or 105000/ 4100		2		2	2	1
		(ii)		all from {secondary oocyte/ female/ mother/ egg}(1)		1		1		
		(iii)		✓ ✓ ✓ x 4 correct (2), 2/3 correct (1)		2		2		
		(iv)		{Nucleus/ DNA / nuclear material/ genetic information} from mother/patient (1) {Nucleus/ DNA/ sperm} from father (1) <u>Mitochondria</u> (DNA) from Donor (1)			3	3		
	(b)	(i)		Move (from one square to another)(1)			1	1		1
		(ii)	I	10mm × 10mm × 10mm = 1000/ 1 x 10 ³ (mm ³)						
			II	40 x 1 000 000 = 40 000 000/ 4 x 10 ⁷ (1)		3		3	3	3
			III	400 000 x 100 = 4 000 000 000 / 4 x 10 ⁹ (1) ECF from II						
				Question 3 total		8	4	12	5	5